COUNTEREXAMPLES TO A CONJECTURE ON 5-REGULAR RICCI-FLAT GRAPHS

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Abstract. Lei and Bai conjectured that every 5-regular graph which is Ricci-flat with respect to Lin–Lu–Yau curvature is either isomorphic to their 72-vertex graph RF^5_{72} or is of Cartesian product type. We disprove this dichotomy in the ordinary graph-theoretic Cartesian-product sense. First, we give an explicit 32-vertex counterexample obtained from two cyclic layers over $\mathbb{Z}/16\mathbb{Z}$. Second, we construct an infinite family $\{H_q : q \geq 5\}$ of 5-regular Ricci-flat graphs which are not ordinary Cartesian products. The graphs H_q are twisted cyclic Petersen bundles: after deleting two distinguished unit cycles one obtains q Petersen fibers, and the monodromy around the base cycle is a nontrivial rotation of the Petersen graph. The curvature verification is reduced to five optimal-assignment matrices, each with a matching cost and a dual lower-bound certificate equal to 6.

1. Introduction

Ollivier's Ricci curvature for Markov chains on metric spaces is defined by optimal transport [9]. In graph theory, Lin, Lu, and Yau introduced a modified idleness-free curvature by taking the derivative of Ollivier curvature at idleness one [7]. This curvature assigns a number $\kappa(x, y)$ to every edge $x \sim y$ of a locally finite graph. A graph is called *Ricci-flat* if

$$\kappa(x, y) = 0$$

for every edge $x \sim y$.

The classification of Ricci-flat graphs is known in several low-degree or high-girth regimes. Lin, Lu, and Yau studied Ricci-flat graphs with girth at least five [8]. Cushing, Kangaslampi, Lin, Liu, Lu, and Yau later corrected and refined part of the cubic girth-five classification [3]. Bai, Lu, and Yau classified Ricci-flat graphs of maximum degree at most four [1]. Degree five is the first regular degree outside the maximum-degree-four classification.

Lei and Bai constructed a 5-regular symmetric Ricci-flat graph RF^5_{72} on 72 vertices and proved that every 5-regular symmetric Ricci-flat graph is

2020 *Mathematics Subject Classification.* 05C75, 05C12, 05C25, 53C21.

Key words and phrases. Lin–Lu–Yau curvature; Ollivier–Ricci curvature; Ricci-flat graph; regular graph; Petersen graph; graph bundle; Cartesian product.

Corresponding author: Guangfu Wang. This work was supported by the Natural Science Foundation of Shandong Province (Grant No. ZR2024MA073).

isomorphic to it [6]. They further conjectured that a general 5-regular Ricci-flat graph is either isomorphic to RF^5_{72} or is of Cartesian product type. This conjecture is also quoted in Hehl's recent work on Lin–Lu–Yau curvature and regular bone-idle graphs [5]. In the present paper, *Cartesian product type* means decomposability as a nontrivial ordinary Cartesian product of finite simple graphs.

Our main result is the following.

Main theorem. *There is a 32-vertex 5-regular Lin–Lu–Yau Ricci-flat graph that is neither isomorphic to RF^5_{72} nor a nontrivial ordinary Cartesian product. Moreover, there are infinitely many such graphs.*

We prove this result in two complementary ways. Section 3 gives a 32-vertex counterexample. Section 4 gives an infinite family of counterexamples. The infinite family is useful because it shows that the failure is structural, not an isolated small-order accident. The construction is a twisted cyclic Petersen bundle. Locally, the five edge types have the same optimal-assignment costs as product-like Ricci-flat configurations; globally, a nontrivial monodromy prevents the graph from being an ordinary Cartesian product.

This also clarifies the relation with the odd-degree bone-idle problem discussed by Hehl. Hehl observes that the Lei–Bai conjecture, if true, would rule out 5-regular bone-idle graphs by combining the Cartesian-product curvature formula with the nonexistence of 3-regular bone-idle graphs [5]. Since the ordinary-product form of the conjecture is false, that route is unavailable. The examples in this paper are Ricci-flat but not bone-idle, so the existence of regular bone-idle graphs of odd degree remains open.

2. Preliminaries

All graphs in this paper are finite, simple, connected, and unweighted. For a vertex x, let $S_1(x)$ be the set of neighbors of x and let $B_1(x) = \{x\} \cup S_1(x)$. For an edge $x \sim y$ whose endpoints have common degree d, define

$$L_{xy} = S_1(x) \setminus B_1(y), \qquad R_{xy} = S_1(y) \setminus B_1(x).$$

The sets L_{xy} and R_{xy} have the same cardinality. Let $\mathcal{A}_{xy}$ be the set of all bijections $\varphi : L_{xy} \to R_{xy}$, and put

$$m(x,y) = \min_{\varphi \in \mathcal{A}_{xy}} \sum_{u \in L_{xy}} \mathrm{d}(u, \varphi(u)).$$

For regular graphs, the Lin–Lu–Yau curvature is computed by the optimal-assignment formula

$$\kappa(x,y) = \frac{1}{d}\big(d + 1 - m(x,y)\big). \tag{1}$$

This form appears in Hehl's work on regular graphs [4]; equivalent matching formulations follow from the optimal-transport definition of Lin–Lu–Yau curvature. In particular, for a 5-regular graph,

$$\kappa(x,y) = 0 \quad \Longleftrightarrow \quad m(x,y) = 6. \tag{2}$$

We also use the Cartesian-product curvature formula. If X is d_X-regular and Y is d_Y-regular, then for an edge $(x_1, y) \sim (x_2, y)$ in $X \square Y$,

$$\kappa^{X\square Y}\big((x_1,y),(x_2,y)\big) = \frac{d_X}{d_X + d_Y}\,\kappa^X(x_1,x_2), \tag{3}$$

and the analogous formula holds for edges in the Y-direction. This formula was proved by Lin, Lu, and Yau [7]; see also the product viewpoint in [2].

3. A 32-vertex counterexample

All subscripts in this section are read in $\mathbb{Z}/16\mathbb{Z}$. Define a graph G by

$$V(G) = \{a_i, b_i : i \in \mathbb{Z}/16\mathbb{Z}\},$$

with adjacency relations

$$a_i \sim a_{i\pm1},\ a_{i\pm6},\ b_i, \qquad b_i \sim b_{i\pm1},\ b_{i\pm7},\ a_i. \tag{4}$$

Equivalently, G consists of two circulant layers with connection sets $\{\pm1, \pm6\}$ and $\{\pm1, \pm7\}$, together with the perfect matching $a_i \sim b_i$.

Lemma 3.1. *The graph G is connected, 5-regular, and has 32 vertices.*

Proof. The displayed neighborhood formula gives degree five at every vertex. The vertex count is $2 \cdot 16 = 32$. The generator 1 occurs in each cyclic layer, so each layer is connected. The matching edges $a_i \sim b_i$ connect the two layers. Hence G is connected. □

The translations

$$\tau_t(a_i) = a_{i+t}, \qquad \tau_t(b_i) = b_{i+t} \qquad (t \in \mathbb{Z}/16\mathbb{Z})$$

are automorphisms of G. Therefore every edge is translation-equivalent to one of

$$a_0a_1, \quad a_0a_6, \quad b_0b_1, \quad b_0b_7, \quad a_0b_0. \tag{5}$$

Theorem 3.2. *The graph G is Ricci-flat with respect to Lin–Lu–Yau curvature.*

Proof. By Lemma 3.1, G is 5-regular. It is enough to compute $m(x, y)$ for the five representative edges in (5). Table 1 lists the sets L_{xy} and R_{xy}. Table 2 gives the distance matrices between them, with rows and columns ordered as in Table 1. These entries are obtained directly from the adjacency lists in (4). No entry can exceed 3, because $u - x - y - v$ is always a path of length 3 from a row vertex $u \in L_{xy}$ to a column vertex $v \in R_{xy}$.

For each matrix $D = (D_{ij})$, Table 3 gives an optimal assignment and a dual lower-bound certificate. The permutation π means that row i is matched to column $\pi(i)$. The vectors $\alpha = (\alpha_i)$ and $\beta = (\beta_j)$ satisfy $\alpha_i + \beta_j \le D_{ij}$ for all i, j, and $\sum_i \alpha_i + \sum_j \beta_j = 6$. Hence every assignment has cost at least 6, while the displayed permutation has cost 6.

Thus $m(x, y) = 6$ for each representative edge. By (2), all five representative edges have Lin–Lu–Yau curvature zero. Translation invariance then gives $\kappa(x, y) = 0$ for every edge of G. □

TABLE 1. The sets $L_{xy} = S_1(x) \setminus B_1(y)$ and $R_{xy} = S_1(y) \setminus B_1(x)$ for the five representative edges of G.

edge xy	L_{xy}	R_{xy}
a_0a_1	$(a_6, a_{10}, a_{15}, b_0)$	(a_2, a_7, a_{11}, b_1)
a_0a_6	$(a_1, a_{10}, a_{15}, b_0)$	(a_5, a_7, a_{12}, b_6)
b_0b_1	(a_0, b_7, b_9, b_{15})	(a_1, b_2, b_8, b_{10})
b_0b_7	(a_0, b_1, b_9, b_{15})	(a_7, b_6, b_8, b_{14})
a_0b_0	$(a_1, a_6, a_{10}, a_{15})$	(b_1, b_7, b_9, b_{15})

TABLE 2. Distance matrices for the five representative edges of G. Rows and columns are ordered as in Table 1.

edge xy	distance matrix
a_0a_1	$\begin{pmatrix} 2 & 1 & 2 & 3 \\ 3 & 3 & 1 & 2 \\ 3 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 \end{pmatrix}$
a_0a_6	$\begin{pmatrix} 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 3 \\ 1 & 3 & 3 & 2 \\ 3 & 2 & 3 & 2 \end{pmatrix}$
b_0b_1	$\begin{pmatrix} 1 & 3 & 3 & 2 \\ 2 & 3 & 1 & 3 \\ 3 & 1 & 1 & 1 \\ 3 & 3 & 1 & 3 \end{pmatrix}$
b_0b_7	$\begin{pmatrix} 2 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 1 & 1 & 1 \end{pmatrix}$
a_0b_0	$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 3 & 2 & 3 & 2 \\ 2 & 3 & 2 & 3 \\ 3 & 3 & 2 & 1 \end{pmatrix}$

TABLE 3. Optimal-assignment certificates for the matrices in Table 2.

edge xy	π	distances along π	$(\alpha; \beta)$
a_0a_1	$(2, 3, 1, 4)$	$(1, 1, 3, 1)$	$((0, 0, 1, 0); (2, 1, 1, 1))$
a_0a_6	$(2, 3, 1, 4)$	$(1, 2, 1, 2)$	$((0, 0, 0, 0); (1, 1, 2, 2))$
b_0b_1	$(1, 2, 4, 3)$	$(1, 3, 1, 1)$	$((0, 1, -1, 1); (1, 2, 0, 2))$
b_0b_7	$(2, 1, 3, 4)$	$(2, 2, 1, 1)$	$((1, 1, 1, 0); (1, 1, 0, 1))$
a_0b_0	$(1, 2, 3, 4)$	$(1, 2, 2, 1)$	$((0, 0, 0, 0); (1, 2, 2, 1))$

Theorem 3.3. *The graph G is not edge-transitive, is not isomorphic to RF^5_{72}, and is not a nontrivial ordinary Cartesian product of graphs. Consequently, G is a counterexample to Conjecture 1 of Lei and Bai under the ordinary Cartesian-product interpretation.*

Proof. First, G is not edge-transitive. The edge a_0a_1 is contained in three distinct 4-cycles, corresponding to the adjacencies a_6a_7, $a_{10}a_{11}$, and b_0b_1 between $L_{a_0a_1}$ and $R_{a_0a_1}$. The edge a_0a_6 is contained in two distinct 4-cycles, corresponding to the adjacencies a_1a_7 and $a_{15}a_5$ between $L_{a_0a_6}$ and $R_{a_0a_6}$. Hence G is not symmetric in the edge-transitive sense.

The graph RF^5_{72} has 72 vertices, whereas G has 32 vertices by Lemma 3.1. Hence $G \ncong RF^5_{72}$.

It remains to exclude nontrivial ordinary Cartesian products. Suppose, for a contradiction, that

$$G \cong X_1 \square \cdots \square X_s$$

with all factors X_j connected and nontrivial. Since the degree in a Cartesian product is the sum of the degrees in the factors, and since G is 5-regular, each factor X_j is regular: fixing all other coordinates and varying one coordinate along paths in X_j forces the degree function of X_j to be constant. Let d_j be the positive degree of X_j. Then $\sum_j d_j = 5$.

By the Cartesian-product curvature formula (3), the Ricci-flatness of G implies that every positive-degree factor is Ricci-flat. No factor can have degree 1, because a connected 1-regular simple graph is K_2, and formula (1) gives $\kappa(K_2) = 2$ on its unique edge. Hence every $d_j \geq 2$. The only way to write 5 as a sum of integers at least 2 is $5 = 2 + 3$. Therefore a nontrivial product decomposition would have the form

$$G \cong X \square Y,$$

where X is a finite connected 2-regular Ricci-flat graph and Y is a finite connected 3-regular Ricci-flat graph.

The graph X must be a cycle C_m. A cycle is Lin–Lu–Yau Ricci-flat precisely when $m \geq 6$: for $m \geq 4$, formula (1) reduces the curvature of an edge to the distance between the two unmatched neighbors, which equals 3 exactly for $m \geq 6$, while C_3 has positive curvature. Since m divides $|V(G)| = 32$, the only possibilities are

$$m = 8, \quad 16, \quad 32.$$

The corresponding orders of the 3-regular factor Y are $4, 2, 1$. Orders 1 and 2 cannot support a simple 3-regular graph. On 4 vertices the only simple 3-regular graph is K_4, and formula (1) gives $\kappa(K_4) = 4/3$, so it is not Ricci-flat. This contradiction proves that G is not a nontrivial ordinary Cartesian product. □

4. An infinite twisted Petersen-bundle family

The 32-vertex example disproves the ordinary-product version of the conjecture. We now show that the failure is infinite and has a transparent structural mechanism.

Let $q \geq 5$ and put $n = 5q$. All subscripts in this section are read in $\mathbb{Z}/n\mathbb{Z}$. Define H_q by

$$V(H_q) = \{a_i, b_i : i \in \mathbb{Z}/n\mathbb{Z}\},$$

with adjacency relations

$$a_i \sim a_{i\pm1},\ a_{i\pm q},\ b_i, \qquad b_i \sim b_{i\pm1},\ b_{i\pm 2q},\ a_i. \tag{6}$$

Thus the first layer has connection set $\{\pm1, \pm q\}$, the second layer has connection set $\{\pm1, \pm 2q\}$, and the two layers are joined by the matching $a_i \sim b_i$.

Lemma 4.1. *For every $q \geq 5$, the graph H_q is connected, 5-regular, and has $10q$ vertices.*

Proof. The connection steps in (6) are distinct for $q \geq 5$, so every vertex has degree five. The vertex count is $2n = 10q$. The unit step 1 connects each cyclic layer, and the matching edges connect the two layers. □

The translations $a_i \mapsto a_{i+t}$, $b_i \mapsto b_{i+t}$ are automorphisms. Hence every edge is translation-equivalent to one of

$$a_0a_1, \quad a_0a_q, \quad b_0b_1, \quad b_0b_{2q}, \quad a_0b_0. \tag{7}$$

Theorem 4.2. *For every $q \geq 5$, the graph H_q is Ricci-flat with respect to Lin–Lu–Yau curvature.*

Proof. By Lemma 4.1, H_q is 5-regular. We compute the five edge types in (7). Table 4 gives the sets L_{xy} and R_{xy}, and Table 5 gives the corresponding distance matrices.

The distance entries in Table 5 are independent of q for $q \geq 5$. To see this systematically, put

$$A = \{\pm1, \pm q\}, \qquad B = \{\pm1, \pm 2q\} \subseteq \mathbb{Z}/5q\mathbb{Z}.$$

For two a-vertices, a path of length at most 2 has index displacement in $A \cup (A+A) \cup \{0\}$; for two b-vertices, it has displacement in $B \cup (B+B) \cup \{0\}$; and for one a-vertex and one b-vertex, it has displacement in $\{0\} \cup A \cup B$. These three elementary criteria determine whether an entry in the table is 1, 2, or at least 3. Since every entry is at most 3 by the path $u - x - y - v$, the table follows by checking the listed displacements modulo $5q$.

Table 6 gives, for each matrix, a permutation of cost 6 and dual vectors α, β with $\alpha_i + \beta_j \leq D_{ij}$ and $\sum_i \alpha_i + \sum_j \beta_j = 6$. Therefore the minimum assignment cost is 6 for every representative edge.

Hence $m(x, y) = 6$ for the five representative edges. By (2) and translation invariance, $\kappa(x, y) = 0$ for every edge of H_q. □

TABLE 4. The sets L_{xy} and R_{xy} for the five representative edges of H_q.

edge xy	L_{xy}	R_{xy}
a_0a_1	$(a_{-1}, a_{-q}, a_q, b_0)$	$(a_2, a_{1-q}, a_{1+q}, b_1)$
a_0a_q	$(a_{-1}, a_1, a_{-q}, b_0)$	$(a_{2q}, a_{q-1}, a_{q+1}, b_q)$
b_0b_1	$(a_0, b_{-1}, b_{-2q}, b_{2q})$	$(a_1, b_2, b_{1-2q}, b_{1+2q})$
b_0b_{2q}	$(a_0, b_{-1}, b_1, b_{-2q})$	$(b_{2q-1}, b_{2q+1}, a_{2q}, b_{4q})$
a_0b_0	$(a_{-1}, a_1, a_{-q}, a_q)$	$(b_{-1}, b_1, b_{-2q}, b_{2q})$

TABLE 5. Distance matrices for the five representative edges of H_q. Rows and columns are ordered as in Table 4.

edge xy	distance matrix
a_0a_1	$\begin{pmatrix} 3 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 1 \end{pmatrix}$
a_0a_q	$\begin{pmatrix} 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 \\ 2 & 3 & 3 & 2 \end{pmatrix}$
b_0b_1	$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 1 \end{pmatrix}$
b_0b_{2q}	$\begin{pmatrix} 3 & 3 & 2 & 2 \\ 1 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 2 & 2 \end{pmatrix}$
a_0b_0	$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 2 & 2 \\ 3 & 3 & 2 & 2 \end{pmatrix}$

TABLE 6. Optimal-assignment certificates for the matrices in Table 5.

edge xy	π	distances along π	$(\alpha; \beta)$
a_0a_1	$(1, 2, 3, 4)$	$(3, 1, 1, 1)$	$((2, 0, 0, 0); (1, 1, 1, 1))$
a_0a_q	$(2, 3, 1, 4)$	$(1, 1, 2, 2)$	$((0, 0, 0, 0); (2, 1, 1, 2))$
b_0b_1	$(1, 2, 3, 4)$	$(1, 3, 1, 1)$	$((0, 2, 0, 0); (1, 1, 1, 1))$
b_0b_{2q}	$(3, 1, 2, 4)$	$(2, 1, 1, 2)$	$((0, 0, 0, 0); (1, 1, 2, 2))$
a_0b_0	$(1, 2, 3, 4)$	$(1, 1, 2, 2)$	$((0, 0, 0, 0); (1, 1, 2, 2))$

Lemma 4.3. *Let*

$$U_q = \{a_i a_{i+1},\ b_i b_{i+1} : i \in \mathbb{Z}/5q\mathbb{Z}\}$$

be the set of unit edges of H_q. Then no edge in U_q lies in a 5-cycle, while every edge outside U_q lies in a 5-cycle. Moreover, deleting U_q leaves q connected components, each isomorphic to the Petersen graph.

Proof. Every non-unit edge is visibly contained in a 5-cycle. For example,

$$a_i, a_{i+q}, a_{i+2q}, a_{i+3q}, a_{i+4q}, a_i$$

contains the a-layer long edges,

$$b_i, b_{i+2q}, b_{i+4q}, b_{i+q}, b_{i+3q}, b_i$$

contains the b-layer long edges, and

$$a_i, b_i, b_{i+2q}, a_{i+2q}, a_{i+q}, a_i$$

contains the matching edge $a_i b_i$.

It remains to show that unit edges are not contained in 5-cycles. By translation, consider a_0a_1 and b_0b_1. A 5-cycle containing a_0a_1 would give a length-four path from a_0 to a_1 avoiding the edge a_0a_1. Such a path uses 0, 2, or 4 matching edges. If it uses no matching edge, write the index displacement as $tq + u$, where tq is the signed contribution of the $\pm q$ steps and u is the signed contribution of the unit steps. Since $|tq + u| < 5q$, the congruence $tq + u \equiv 1 \pmod{5q}$ implies $tq + u = 1$. If $t \neq 0$, then $|u| = |1 - tq| \geq q - 1$, whereas the presence of a $\pm q$ step gives $|u| \leq 3$; this is impossible for $q \geq 5$. If $t = 0$, then the number of $\pm q$ steps is even, so the number of unit steps is even and u is even; again $u \neq 1$. If the path uses two matching edges, it has only two horizontal steps, whose possible sums from $\{\pm 1, \pm q, \pm 2q\}$ cannot equal 1 modulo $5q$; four matching edges give zero displacement. Hence a_0a_1 lies in no 5-cycle.

The proof for b_0b_1 is analogous, but the no-matching case uses steps from $\{\pm 1, \pm 2q\}$. Write the displacement as $2tq + u$, where u is the unit-step contribution. The equation is $2tq + u = 1 + 5q\ell$. Reducing modulo q gives $u \equiv 1 \pmod{q}$. For $q \geq 6$, the bound $|u| \leq 4$ forces $u = 1$; then the number of unit steps is odd, so the number of $\pm 2q$ steps and the integer t are odd, while $2t = 5\ell$ is impossible. For $q = 5$, the only possibilities are $u = 1$ and $u = -4$; the first is impossible by the same equation, and the second would require four unit steps and gives displacement -4, not 1 modulo 25. The cases with two or four matching edges are as above.

Finally, after deleting U_q, the residue class of the index modulo q is preserved. For a fixed residue $r \in \mathbb{Z}/q\mathbb{Z}$, the component on

$$\{a_{r+tq}, b_{r+tq} : t \in \mathbb{Z}/5\mathbb{Z}\}$$

has edges

$$a_t \sim a_{t\pm 1}, \qquad b_t \sim b_{t\pm 2}, \qquad a_t \sim b_t \quad (t \in \mathbb{Z}/5\mathbb{Z}),$$

a standard presentation of the Petersen graph. Thus deleting U_q leaves q Petersen components. $\square$

Theorem 4.4. *For every $q \geq 5$, the graph H_q is not a nontrivial ordinary Cartesian product of graphs. Consequently, the graphs H_q form an infinite family of non-product 5-regular Ricci-flat counterexamples to the Lei–Bai conjectural dichotomy under the ordinary Cartesian-product interpretation.*

Proof. Assume that H_q is a nontrivial ordinary Cartesian product. As in the proof of Theorem 3.3, the product curvature formula forces every positive-degree factor to be Ricci-flat, and no factor can have degree 1. Since H_q is 5-regular, the degree split must be $2+3$. Hence

$$H_q \cong C_m \square Y,$$

where $m \geq 6$ and Y is a finite connected 3-regular Ricci-flat graph.

The graph Y is triangle-free. Indeed, if a 3-regular Ricci-flat graph had an edge xy with a common neighbor, then $|S_1(x) \setminus B_1(y)| \leq 1$ and formula (1) would give $\kappa(x, y) > 0$, a contradiction.

We claim that no C_m-direction edge in $C_m \square Y$ lies in a 5-cycle. A 5-cycle in the product projects to closed walks in both factors. Since $m \geq 6$, its projection to C_m cannot be a nontrivial closed walk of odd length at most five. A 5-cycle containing a C_m-direction edge must therefore use either two or four C_m-direction edges. If it used two C_m-direction edges, then the remaining three edges would project to a closed walk of length three in Y, which would be a triangle. If it used four C_m-direction edges, the projection to Y would have length one, impossible. Therefore any edge of $C_m \square Y$ that lies in a 5-cycle must be a Y-direction edge.

By Lemma 4.3, all edges outside U_q lie in 5-cycles. Hence all edges outside U_q must be Y-direction edges. There are $3|V(H_q)|/2 = 15q$ Y-direction edges in $C_m \square Y$, and there are exactly $15q$ edges outside U_q in H_q. Thus the Y-direction edges are precisely the non-unit edges, and the C_m-direction edges are precisely the unit edges U_q.

But the subgraph formed by the C_m-direction edges in a Cartesian product is a disjoint union of $|V(Y)|$ cycles, each of length m. In H_q, the unit-edge subgraph is exactly the disjoint union of the two cycles

$$a_0 a_1 \cdots a_{5q-1} a_0, \qquad b_0 b_1 \cdots b_{5q-1} b_0.$$

Therefore $|V(Y)| = 2$, which is impossible for a simple 3-regular graph. This contradiction proves that H_q is not an ordinary Cartesian product.

Since $|V(H_q)| = 10q \neq 72$, none of the graphs H_q is isomorphic to RF^5_{72}. Together with Theorem 4.2, this gives infinitely many counterexamples. $\square$

Remark 4.5. The mechanism behind H_q is a nontrivial cyclic Petersen bundle. Write each index uniquely as $i = r + tq$ with $r \in \mathbb{Z}/q\mathbb{Z}$ and $t \in \mathbb{Z}/5\mathbb{Z}$. For fixed r, the non-unit edges induce a Petersen graph on the ten vertices a_{r+tq}, b_{r+tq}. Unit edges connect the fiber over r to the fiber over $r+1$. For $r \neq q-1$ this connection preserves t; from $r = q-1$ to $r = 0$ it sends t to

$t+1$. Thus the monodromy around the base cycle is the nontrivial order-five rotation of the Petersen graph. The ordinary product would have trivial monodromy. The curvature calculation is local and survives this twist; the global Cartesian product structure does not.

5. Bone-idleness and minimality issues

The examples above are Ricci-flat but not bone-idle. For the 32-vertex graph G, the zero-idleness assignment formula gives

$$\kappa_0 = 0 \quad \text{on the edges } a_i a_{i+1} \text{ and } b_i b_{i+1},$$

and

$$\kappa_0 = -\frac{1}{5} \quad \text{on the edges } a_i a_{i+6},\ b_i b_{i+7},\ \text{and } a_i b_i.$$

For the infinite family H_q, the same calculation gives

$$\kappa_0 = 0 \quad \text{on the unit edges } a_i a_{i+1} \text{ and } b_i b_{i+1},$$

and

$$\kappa_0 = -\frac{1}{5} \quad \text{on the long-layer and matching edges.}$$

Therefore these constructions do not produce 5-regular bone-idle graphs.

The present paper does not claim that the 32-vertex graph is the smallest possible counterexample among all 5-regular Ricci-flat graphs. Proving such a global minimality statement would require either a structural theorem excluding all non-product examples on fewer than 32 vertices or a reproducible exhaustive enumeration with independent certificates. What is proved here is stronger in a different direction: there are infinitely many non-symmetric, non-product 5-regular Ricci-flat graphs. A natural next problem is to combine the local assignment constraints used above with canonical generation of 5-regular graphs in order to settle the minimum-order question.

Acknowledgments

The authors thank the mathematical community working on discrete Ricci curvature for the background results on which this note builds.

Funding

This work was supported by the Natural Science Foundation of Shandong Province (Grant No. ZR2024MA073).

Data availability statement

No datasets were generated or analysed during the current study. A short verification script for the finite assignment calculations is included with the submission files for editorial convenience.

Conflict of interest statement

The authors declare that they have no conflict of interest.

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